

ECE752 - Project Progress Report

3 April 2025

Project title: Adaptive Demand-Based Caching with Effective Address Filtering (ADC-EAF)**Project Members:** Dhruv Singhi, Yash A Deshpande, Austin James**What we have accomplished so far:**

Successfully set up and built the following SPEC2017 benchmarks to analyze baseline performance across different access patterns: **perlbench, gcc, mcf, omnetpp, xalancbmk, x264, deepsjeng, leela, exchange2, xz, specrand**. Understood Intel PIN's instrumentation technique and tested default Memory analysis Pintools with custom programs, built-in Linux utilities and selected benchmarks.

We identified unexpected behavior in Intel PIN's built-in cache subsystem, which reported anomalous miss rates. This led us to modify and extend the existing base d-cache to include an LLC L2 implementation.

Our design requires a **Least Recently Used (LRU)** cache replacement policy as the baseline. Since Intel PIN lacks a built-in cache class for LRU, we implemented our own using 2 approaches: 'counter-based' & 'doubly linked list-based'. We also evaluated our LRU policy against the built-in Round Robin and Direct Mapped policies

Table1: Net average cache access performance across the 3 replacement policies taken together

Benchmark Group	L1 Load Hits	L1 Store Hits	L2 Load Hits	L2 Store Hits
Sequential (perl, gcc, xalanc, x264)	95.79%	98.07%	63.85%	61.14%
Random (605 MCF)	80.58%	95.34%	17.68%	21.99%

Table1 verifies our expected observation of sequential benchmarks having high L1 hit rates (~95–98%) and moderate L2 hit rates (~60%), indicating strong locality. Random access benchmark has much lower hit rates (L1 ~80%, L2 ~18–22%), showing poor locality. LRU is the best for sequential workloads, maintaining high hit rates by keeping recently used data longer. Direct Mapped suffers from conflict misses, especially in workloads with large working sets. Round Robin performed the worst, particularly for random workloads, as it blindly cycles through cache slots.

What is in Progress and remains to be accomplished:

We have accomplished our objective of setting up the foundational design to build our ideas on. Additionally, we have also implemented the algorithm for the V-way set-associative cache of which we are currently in the phase of debugging identified issues. Post which we will work on integrating an evicted address filter to verify our expected results with the Benchmark suite we have selected.